

Relation Between Fermi–Amaldi and Exact Exchange Energy Densities

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I. MAIN TEXT

The goal is to establish a relationship between the exact exchange energy density (closed-shell):

$$e_x^{\text{exact}}(\mathbf{r}) = -\frac{1}{4} \int \frac{|\gamma(\mathbf{r}; \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}', \quad (1)$$

where

$$\gamma(\mathbf{r}, \mathbf{r}') = 2 \sum_{i=1}^{N/2} \varphi_i^*(\mathbf{r}) \varphi_i(\mathbf{r}') \quad (2)$$

is the one-electron reduced density matrix [1, 2], and the Fermi–Amaldi [3, 4] energy density expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N ,

$$e_x^{\text{FA}}(\mathbf{r}) = -\frac{1}{2N} \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \quad (3)$$

When comparing two integrals with different integrands $f(x)$ and $g(x)$, on the interval $[a, b]$, the following is true

$$\int_a^b f(x) dx \geq \int_a^b g(x) dx, \quad \text{when } f(x) \geq g(x). \quad (4)$$

Both integrands in Eqs. (1) and (3) share the same denominator, $1/|\mathbf{r} - \mathbf{r}'|$, which is positive everywhere in space. Therefore, by the Cauchy–Schwarz inequality:

$$\rho(\mathbf{r})\rho(\mathbf{r}') \geq |\gamma(\mathbf{r}, \mathbf{r}')|^2 \quad \text{for all } \mathbf{r}, \mathbf{r}', \quad (5)$$

we obtain the following relationship:

$$\int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' \geq \int \frac{|\gamma(\mathbf{r}, \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' \quad \text{for all } \mathbf{r}. \quad (6)$$

Then, by the definitions (1) and (3):

$$\frac{e_x^{\text{FA}}(\mathbf{r})}{e_x^{\text{exact}}(\mathbf{r})} \geq \frac{2}{N} \quad \text{for all } \mathbf{r}. \quad (7)$$

Thus, the lower bound of the relationship between the two closed-shell exchange energy densities is given by a function of the electron number. Note, that for $N = 2$, the lower bound is a unity [5–7].

In the asymptotic limit, the denominator of Eqs. (1) and (3) can be approximated as $|\mathbf{r} - \mathbf{r}'| \approx |\mathbf{r}|$, revealing that the exchange energy densities share the same asymptotic behavior:

$$e_x^{\text{exact}}(\mathbf{r}) \sim e_x^{\text{FA}}(\mathbf{r}) \sim -\frac{\rho(\mathbf{r})}{2|\mathbf{r}|}, \quad (8)$$

$$\lim_{|\mathbf{r}| \rightarrow \infty} \frac{e_x^{\text{FA}}(\mathbf{r})}{e_x^{\text{exact}}(\mathbf{r})} = 1. \quad (9)$$

Conversely, in the limit of infinitesimally small interelectronic distances ($\mathbf{r} \rightarrow \mathbf{r}'$), the spatial parts of both integrands approach equality:

$$|\gamma(\mathbf{r}, \mathbf{r}')|^2 \rightarrow \rho(\mathbf{r})\rho(\mathbf{r}') \quad \text{as } \mathbf{r} \rightarrow \mathbf{r}'. \quad (10)$$

Because the exact exchange prefactor is larger than the Fermi–Amaldi prefactor for $N > 2$, the region of small interelectronic distances is dominated by the exact exchange energy density:

$$\frac{e_x^{\text{FA}}(\mathbf{r})}{e_x^{\text{exact}}(\mathbf{r})} \leq 1 \quad \text{as } \mathbf{r} \rightarrow \mathbf{r}'. \quad (11)$$

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